

Cauchy-Euler Equation.

□ Non-Homogeneous LDE with variable coefficients.

An equation of the following form:-

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n y = F(x).$$

OR,

$$a_0 x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + a_2 x^{n-2} y^{(n-2)} + \dots + a_n y = F(x).$$

is called non-homogeneous LDE with variable coefficients.

Euler Equation:- $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Procedure: Let, $x = e^t$

$$\therefore \ln x = \ln e^t = t$$

$$\therefore t = \ln x.$$

$$\therefore \frac{dt}{dx} = \frac{1}{x}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= \frac{dy}{dt} \cdot \frac{1}{x}$$

$$\therefore x \frac{dy}{dx} = Dy \left[\text{Let, } D = \frac{d}{dt} \right]$$

OPDE Cauchy-Euler Equation.

Find the general solution of each of the following differential equation.

1. $x^2 y'' + xy' + 4y = 2x \ln x$

2. $x^2 y'' - 4xy' + 4y = 4x^2 - 6x^3, y(2) = 4$
 $y'(2) = -1$

3. $x^3 y''' - x^2 y'' + 2xy' - 2y = x^3$

4. $x^3 y'' + xy' + y = 4 \sin(\ln x)$

5. $x^3 \frac{d^3 y}{dx^3} - 4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} = 4 \ln x$

Soln-1:

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 4y = 2x \ln x \quad (1)$$

$$\text{Let, } x = e^t$$

$$\therefore \ln x = \ln e^t \quad [\text{both side } \ln]$$

$$\therefore t = \ln x$$

$$\therefore \frac{dt}{dx} = \frac{1}{x}$$

$$\text{Now, } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x} \cdot \frac{dy}{dt}$$

$$\therefore x \frac{dy}{dx} = Dy \quad [\text{Let, } D = \frac{d}{dt}]$$

Similarly,

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

put these value in eqn (1),

$$\therefore D(D-1)y + Dy + 4y = 2e^t \cdot t$$

$$\Rightarrow (D^2 - D)y + Dy + 4y = 2te^t$$

$$\Rightarrow (D^2 - D + D + 4)y = 2 + e^t$$

$$\Rightarrow (D^2 + 4)y = 2 + e^t$$

$$\therefore \frac{d^2 y}{dt^2} + 4y = 2 + e^t \quad \text{--- (2)}$$

(u.c method)

$$\text{Let, } \frac{d^2 y}{dt^2} + 4y = 0 \quad \text{--- (3)}$$

Also let,

$y = e^{mt}$ be a trial soln of (3),

$$\therefore \frac{dy}{dt} = m e^{mt}, \quad \frac{d^2 y}{dt^2} = m^2 e^{mt}$$

put these value in eqn (3),

$$\therefore (m^2 + 4)e^{mt} = 0, \text{ Since } e^{mt} \neq 0$$

auxilliary eqn will be,

$$m^2 + 4 = 0 \Rightarrow m^2 = -4 \therefore m = \pm 2i$$

$$\therefore y_1 = C_1 \sin 2t + C_2 \cos 2t$$

Again from (2)

$$\frac{d}{dx}(uv) = u'v + uv' \quad \parallel \quad \frac{d}{dx}(uv) = uv' + u'v.$$

$$\text{VC set, } S_1 = \{t, 1\}, S_2 = \{e^t\}$$

$$\therefore S = \{te^t, e^t\}$$

$$\therefore y_2 = Ate^t + Be^t$$

$$\therefore y_2' = Ate^t + Ae^t + Be^t$$

$$\therefore y_2'' = Ate^t + Ae^t + Ae^t + Be^t$$

$$= Ate^t + 2Ae^t + Be^t$$

$$\text{put } y_2, y_2'' \text{ in (2)}$$

$$\therefore Ate^t + 2Ae^t + Be^t + 4Ate^t + 4Be^t = 2te^t$$

$$\Rightarrow 5Ate^t + (2A + 5B)e^t = 2te^t$$

Comparing both sides

$$5A = 2 \quad \& \quad 2A + 5B = 0$$

$$\therefore A = \frac{2}{5} \Rightarrow 2 \times \frac{2}{5} + 5B = 0$$

$$\therefore B = -\frac{4}{25}$$

$$\therefore y_2 = \frac{2}{5} te^t - \frac{4}{25} e^t$$

$\therefore$ General soln is $y = y_1 + y_2$

$$\therefore y = c_1 \sin 2t + c_2 \cos 2t + \frac{2}{5} te^t - \frac{4}{25} e^t$$

$$= c_1 \sin 2(\ln x) + c_2 \cos 2(\ln x)$$

$$+ \frac{(2)}{5} x \ln x - \frac{4x}{25}$$

Ans

Soln-4:-

Given,

$$x^2 y'' + xy' + y = 4 \sin(\ln x) \quad \text{--- (1)}$$

Let, $x = e^t$

$$\therefore \ln x = \ln e^t$$

$$\therefore t = \ln x$$

$$\therefore \frac{dt}{dx} = \frac{1}{x}$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\therefore x \frac{dy}{dx} = Dy$$

Similarly,

$$x^2 \frac{d^2y}{dx^2} = D(D-1)y$$

(i) $\Rightarrow$

$$D(D-1)y + Dy + y = 4\sin(t)$$

$$\Rightarrow (D^2 - D + D + 1)y = 4\sin t$$

$$\Rightarrow (D^2 + 1)y = 4\sin t$$

$$\therefore \frac{d^2y}{dt^2} + y = 4\sin t \quad \text{--- (2)}$$

$$\text{Let, } \frac{d^2y}{dt^2} + y = 0 \quad \text{--- (3)}$$

Also let, $y = e^{mt}$

$y' = me^{mt}$ and $y'' = m^2 e^{mt}$
put these value in eqn (3)

$$(m^2 + 1)e^{mt} = 0$$

Since, $e^{mt} \neq 0$

$$\frac{d}{dx}(uv) = uv' + u'v.$$

$$\therefore m^2 + 1 = 0$$

$$\therefore m = \pm i$$

$$\therefore y_1 = C_1 \sin t + C_2 \cos t$$

$$\text{From (1), } S = \{ \sin t, \cos t \}$$

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$$\therefore \text{UC set, } S = \{ + \sin t, + \cos t \}$$

$$\therefore y_2 = A + \sin t + B + \cos t$$

$$\therefore y_1' = A + \cos t + A \sin t + B \cos t - B \sin t$$

$$\therefore y_2'' = -A \sin t + A \cos t + A \cos t - B \sin t$$

$$= 2A \cos t - 2B \sin t - A + \sin t - B + \cos t$$

put these value in eqn (2)

$$\therefore 2A \cos t - 2B \sin t - A + \sin t - B + \cos t + A \sin t$$

$$\Rightarrow -2B \sin t + 2A \cos t = 4 \sin t$$

$$v'u + v'u = (vu) \frac{b}{ab}$$

Comparing both sides, $0 = 1 + \sin t$

$$-2B = 4, \quad 2A = 0$$

$$\therefore B = -2, \quad A = 0$$

(1) $\Rightarrow$

$$D(D-1)y + Dy + y = 4 \sin t$$

$$\Rightarrow (D^2 - D + D + 1)y = 4 \sin t$$

$$\Rightarrow (D^2 + 1)y = 4 \sin t$$

$$\therefore \frac{d^2 y}{dt^2} + y = 4 \sin t \quad (2)$$

$$\text{Let, } \frac{d^2 y}{dt^2} + y = 0 \quad (3)$$

Also

$$\therefore y_2 = 0 - 2 + \cos t$$

$$\therefore \text{General soln in } y = y_1 + y_2$$

$$\therefore y_2 = 0 - 2 + \cos t$$

General soln in $y = y_1 + y_2$

$$\therefore y = C_1 \sin t + C_2 \cos t - 2 + \cos t$$

$$= C_1 \sin(\ln x) + C_2 \cos(\ln x) - 2 \ln x \cdot \cos(\ln x)$$

An

$$\text{iii. } x^2 y'' - 4xy' + 4y = 4x^2 - 6x^3, y(2) = 4$$

Let,

$$x = e^t \quad \text{--- (i) } y'(2) = -1.$$

$$\Rightarrow \ln x = \ln e^t$$

$$\Rightarrow \ln x = t$$

$$\therefore \frac{dt}{dx} = \frac{1}{x}$$

$$\text{Now, } \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{x}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dt}$$

$$\Rightarrow x \frac{dy}{dx} = Dy \quad \left[\text{Let, } D = \frac{d}{dt} \right]$$

Similarly,

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

put these value in eqn (i).

$$D(D-1)y - 4Dy + 4y = 4(e^t)^2 - 6(e^t)^3$$

$$\Rightarrow D^2y - Dy - 4Dy + 4y = 4e^{2t} - 6e^{3t}$$

$$\Rightarrow D^2y - 5Dy + 4y = 4e^{2t} - 6e^{3t}$$

$$\Rightarrow \frac{d^2y}{dt^2} - 5\frac{dy}{dt} + 4y = 4e^{2t} - 6e^{3t} \quad \text{--- (ii)}$$

Let,

$$\frac{d^2y}{dt^2} - 5\frac{dy}{dt} + 4y = 0 \quad \text{--- (iii)}$$

Also Let,

$y = e^{mt}$ be a trial soln of (iii)

$$y' = me^{mt}$$

$$y'' = m^2e^{mt}$$

put these y, y', y'' value in eqn (iii).

$$m^2e^{mt} - 5me^{mt} + 4e^{mt} = 0$$

$$\Rightarrow (m^2 - 5m + 4)e^{mt} = 0$$

Since, $e^{mt} \neq 0$

$$m^2 - 5m + 4 = 0$$

$$\Rightarrow m^2 - 4m - m + 4 = 0$$

$$\Rightarrow m(m-4) - 1(m-4) = 0$$

$$\Rightarrow (m-1)(m-4) = 0$$

$$\therefore m = 1, 4$$

$$\therefore y_1 = c_1 e^t + c_2 e^{4t}$$

Again from eqn (ii),

$$S_1 = \{e^{2t}\}, S_2 = \{e^{3t}\}$$

or set,

$$S = \{e^{2t}, e^{3t}\}$$

$$\therefore y_2 = Ae^{2t} + Be^{3t}$$

$$y_2' = 2Ae^{2t} + 3Be^{3t}$$

$$y_2'' = 4Ae^{2t} + 9Be^{3t}$$

put these y_2, y_2', y_2'' value in eqn (ii).

$$4Ae^{2t} + 9Be^{3t} - 5(2Ae^{2t} + 3Be^{3t}) + 4(Ae^{2t} + Be^{3t}) = 4e^{2t} - 6e^{3t}$$

$$\Rightarrow 4Ae^{2t} + 9Be^{3t} - 10Ae^{2t} - 15Be^{3t} + 4Ae^{2t} + 4Be^{3t} = 4e^{2t} - 6e^{3t}$$

$$\Rightarrow (4A - 10A + 4A)e^{2t} + (9B - 15B + 4B)e^{3t} = 4e^{2t} - 6e^{3t}$$

Comparing both sides,

$$\begin{array}{l|l} 4A - 10A + 4A = 4 & 9B - 15B + 4B = -6 \\ \hline \Rightarrow -2A = 4 & \Rightarrow -2B = -6 \\ \therefore A = -2 & \therefore B = 3 \end{array}$$

$$\therefore Y_2 = -2e^{2t} + 3e^{3t}$$

General soln is, $y = Y_1 + Y_2$

$$y = C_1 e^t + C_2 e^{4t} - 2e^{2t} + 3e^{3t}$$

$$y = C_1 x + C_2 x^4 - 2x^2 + 3x^3 \quad \text{--- (iv)}$$

$$\therefore y' = C_1 + 4C_2 x^3 - 4x + 9x^2 \quad \text{--- (v)}$$

put, $x=2$ and $y=4$ in eqn (iv)

$$4 = C_1 \times 2 + C_2 \times 2^4 - 2 \times 2^2 + 3 \times 2^3$$

$$\Rightarrow 4 = 2C_1 + 16C_2 - 8 + 24$$

$$\Rightarrow 2C_1 + 16C_2 = -12 \text{ ————— (vi)}$$

put, $x=2$ and $y=-1$ in eqn (v),

$$-1 = C_1 + 4C_2 \times 2^3 - 4 \times 2 + 9 \times 2^2.$$

$$\Rightarrow -1 = C_1 + 32C_2 - 8 + 36$$

$$\Rightarrow C_1 + 32C_2 = -29 \text{ ————— (vii)}$$

$$\therefore C_1 = \frac{5}{3}$$

$$\therefore C_2 = -\frac{23}{24}$$

Am

Solution:-03 Given, $x^3 y''' - x^2 y'' + 2xy' - 2y = x^3$ — (i)

Let,

$$x = e^t \Rightarrow \ln x = \ln e^t = t$$

$$\Rightarrow \ln x = \ln e^t = t$$

$$\Rightarrow \ln x = t \Rightarrow \frac{dt}{dx} = \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} \cdot \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{x}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dt}$$

$$\Rightarrow x \frac{dy}{dx} = \frac{dy}{dt}$$

Similarly,

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

(iii)

$$(a^m)^n = a^{mn}$$

Similarly, (revised) so: modified

$$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y$$

put these value in eqn (i)

$$\Rightarrow D(D-1)(D-2)y - D(D-1)y + 2Dy - 2y = (e^t)^3$$

$$\Rightarrow [D(D-1)(D-2) - D(D-1) + 2D - 2]y = e^{3t}$$

$$\Rightarrow [D(D^2 - 2D - D + 2) - D^2 + D + 2D - 2]y = e^{3t}$$

$$\Rightarrow (D^3 - 2D^2 - D^2 + 2D - D^2 + D + 2D - 2)y = e^{3t}$$

$$\Rightarrow (D^3 - 4D^2 + 5D - 2)y = e^{3t}$$

$$\Rightarrow D^3 y - 4D^2 y + 5Dy - 2y = e^{3t}$$

$$\Rightarrow \frac{d^3 y}{dt^3} - 4 \frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} - 2y = e^{3t}$$

(ii)

Let,

$$\frac{d^3 y}{dt^3} - 4 \frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} - 2y = 0$$

(iii)

Also Let,

$$y = e^{mt}$$

$$y' = m e^{mt}$$

$$y'' = m^2 e^{mt}$$

$$y''' = m^3 e^{mt}$$

put these value in y, y', y'', y''' value
in eqn (iii)

$$(m^3 - 4m^2 + 5m - 2) e^{mt} = 0$$

Since, $e^{mt} \neq 0$

$$\Rightarrow m^3 - 4m^2 + 5m - 2 = 0$$

$$m = 1, 2, 1$$

$$y_1 = c_1 e^t + c_2 e^{2t} + c_3 t e^t$$

Again from eqn (ii)

VC set,

$$S = \{e^{3t}\}$$

$$y_2 = Ae^{3t}$$

$$y_2' = 3Ae^{3t}$$

$$y_2'' = 9Ae^{3t}$$

$$y_2''' = 27Ae^{3t}$$

put these y_2, y_2', y_2'', y_2''' value

in eqn (ii)

$$\Rightarrow 27Ae^{3t} = 4(9Ae^{3t}) + 5(3Ae^{3t}) - 2(Ae^{3t}) = e^{3t}$$

$$\Rightarrow 27Ae^{3t} - 36Ae^{3t} + 15Ae^{3t} - 2Ae^{3t} = e^{3t}$$

(ii) nps most niofa

$$\Rightarrow (27A - 36A + 15A - 2A)e^{3t} = e^{3t}$$

$$\Rightarrow (4A - 38A)e^{3t} = e^{3t}$$

$$\Rightarrow 4A = 1$$

$$\therefore A = \frac{1}{4}$$

$$\therefore y_2 = \frac{1}{4}e^{3t}$$

General soln is, $y = y_1 + y_2$

$$y = c_1 e^t + c_2 e^{2t} + c_3 t e^t + \frac{1}{4} e^{3t}$$

$$= c_1 x + c_2 x^2 + c_3 x \ln x + \frac{1}{4} x^3$$

$$\frac{1}{x} \cdot \frac{1}{x} = \frac{1}{x^2}$$

$$\left[\frac{1}{x} = 0 \right] \Rightarrow \frac{1}{x} = 0$$

$$B(1-D)D = \frac{1}{x}$$

Solution:-05 Given, -

$$x^3 \frac{d^3 y}{dx^3} - 4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} = 4 \ln x$$

Let,

$$x = e^t$$

$$\Rightarrow \ln x = \ln e^t$$

$$\Rightarrow t = \ln x$$

$$\Rightarrow \frac{dt}{dx} = \frac{1}{x}$$

Now,

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{x}$$

$$\Rightarrow x \frac{dy}{dx} = Dy \left[\because D = \frac{d}{dt} \right]$$

Similarly,

$$x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

Similarly,

$$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y = 6$$

put these value in eqn (i),

$$D(D-1)(D-2)y - 4D(D-1)y + 8Dy = 4t$$

$$\Rightarrow [(D^2 - D)(D-2) - 4D^2 + 4D + 8D]y = 4t$$

$$\Rightarrow (D^3 - 2D^2 - D^2 + 2D - 4D^2 + 4D + 8D)y = 4t$$

$$\Rightarrow (D^3 - 7D^2 + 14D)y = 4t$$

$$\Rightarrow D^3 y - 7D^2 y + 14Dy = 4t$$

$$\Rightarrow \frac{d^3 y}{dt^3} - 7 \frac{d^2 y}{dt^2} + 14 \frac{dy}{dt} = 4t$$

Let,

$$\frac{d^3 y}{dt^3} - 7 \frac{d^2 y}{dt^2} + 14 \frac{dy}{dt} = 0$$

(iii)

Also Let,

$$y = e^{mt}$$

$$\Rightarrow y' = m e^{mt}$$

$$\Rightarrow y'' = m^2 e^{mt}$$

$$\Rightarrow y''' = m^3 e^{mt}$$

put these y, y', y'', y''' value in
eqn (iii),

$$(m^3 - 7m^2 + 14m) e^{mt} = 0$$

Since, $e^{mt} \neq 0$

$$\Rightarrow m^3 - 7m^2 + 14m = 0$$

$$(i) m = 0, \frac{7}{2} \pm \frac{\sqrt{7}}{2} i$$

$$y_1 = c_1 + e^{\frac{7}{2}t} \left(c_2 \sin \frac{\sqrt{7}}{2} t + c_3 \cos \frac{\sqrt{7}}{2} t \right)$$

(iii)

Again from (ii),

$$S = \{ \pm 1, \pm 1 \} \text{ APL}$$

$$N = A \times 80$$

$$\text{VC set, } S = \{ \pm 1, \pm 1 \} \frac{N}{80} = A \ll$$

$$y_2 = A t^2 + B t \quad \frac{1}{F} = A \therefore$$

$$y_2' = 2A t + B$$

$$y_2'' = 2A$$

$$y_2''' = 0$$

put these y_2, y_2', y_2'', y_2''' value in

eqn. (ii),

$$\Rightarrow 0 - 7(2A) + 14(2A t + B) = 4 t$$

$$\Rightarrow -14A + 28A t + 14B = 4 t$$

$$\Rightarrow (28A) t + (-14A + 14B) = 4 t$$

mA

Comparing the both sides

$$\begin{array}{l|l} 28A = 4 & -14A + 14B = 0 \\ \Rightarrow A = \frac{4}{28} & -14 \times \frac{1}{7} + 14B = 0 \\ \therefore A = \frac{1}{7} & \Rightarrow 14B = \frac{14}{7} \end{array}$$

$$\therefore B = \frac{1}{7}$$

$$\therefore Y_2 = \frac{1}{7} t^2 + \frac{1}{7} t$$

General soln is, $y = Y_1 + Y_2$

$$y = C_1 + e^{\frac{7}{2}t} \left(C_2 \sin \frac{\sqrt{7}}{2} t + C_3 \cos \frac{\sqrt{7}}{2} t \right) + \frac{1}{7} t^2 + \frac{1}{7} t$$

$$\therefore y = C_1 + x^{\frac{7}{2}} \left(C_2 \sin \frac{\sqrt{7}}{2} \ln x + C_3 \cos \frac{\sqrt{7}}{2} \ln x \right) + \frac{1}{7} (\ln x)^2 + \frac{1}{7} \ln x$$

Ans